Project Plan

Prompt

Act as a Software Architect and create a project plan for the Inventory Management System (IMS), considering a team of 4 developers.

Include:

- Major epics and deliverables
- Estimated timelines, dependencies, and milestones
- Key stakeholders and their responsibilities
- Assumptions and team constraints

Present the plan in structured tables, ensuring the timeline and workload are realistic for the team size.

Response

Inventory Management System – Project Plan

1. Project Overview

Objective

Build a web-based Inventory Management System (IMS) that enables:

- Product and inventory tracking
- Real-time stock updates
- Warehouse operations management
- Supplier and purchase management
- Reporting and audit tracking

2. Project Team Structure

Core Development Team (4 Developers)

Role	Responsibility
Tech Lead / Senior Full Stack Developer	Architecture, code reviews, backend leadership
Backend Developer	APIs, business logic, database integration
Frontend Developer	UI/UX implementation
Full Stack Developer	Shared frontend/backend features, integrations

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Additional Stakeholders

Stakeholder	Responsibility
Product Owner	Business decisions and prioritization
Business Analyst	Requirements gathering and documentation
Project Manager / Scrum Master	Sprint planning and delivery tracking
QA Engineer	Testing and quality validation
DevOps Engineer	CI/CD, environments, deployments
Warehouse Manager	Workflow validation
Inventory Operations Team	User acceptance testing
IT/Security Team	Security and infrastructure review

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3. Recommended Delivery Model

Methodology

Agile Scrum

Sprint Duration

2 Weeks

Total Estimated Duration

20 Weeks

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4. Major Project Phases

Phase	Duration
Discovery & Planning	2 Weeks
Architecture & Design	2 Weeks
Core Development	10 Weeks
Integration & QA	3 Weeks
UAT & Stabilization	2 Weeks
Production Deployment	1 Week

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5. Major Epics

Epic 1 – Project Setup & Foundation

Objective

Establish development standards, infrastructure, and project structure.

Features

- Repository setup
- CI/CD pipeline
- Environment setup
- Authentication framework
- Logging framework
- Error handling
- Base UI framework

Estimated Duration

2 Weeks

Assigned Team

- Tech Lead
- Full Stack Developer

Epic 2 – Authentication & User Management

Objective

Provide secure access and role-based permissions.

Features

- Login/logout
- Password reset
- JWT/session management
- Role-based access control

- User management
- Audit logging

Estimated Duration

2 Weeks

Dependencies

Project setup completed

Epic 3 – Product Management

Objective

Manage inventory catalog and master data.

Features

- Product CRUD
- Category management
- SKU generation
- Barcode support
- Product search/filtering

Estimated Duration

2 Weeks

Epic 4 – Inventory Management

Objective

Track inventory quantities and movements.

Features

- Real-time stock tracking
- Multi-warehouse support
- Inventory ledger
- Stock validation
- Inventory history

Estimated Duration

3 Weeks

Epic 5 – Stock Inward Management

Objective

Manage incoming inventory workflows.

Features

- Goods receiving
- Purchase receiving
- Partial receiving
- Damaged goods tracking
- Supplier stock intake

Estimated Duration

2 Weeks

Epic 6 – Stock Outward & Dispatch

Objective

Handle stock deductions and dispatch operations.

Features

- Dispatch workflow
- Stock deduction
- Returns handling
- Transfer management
- Dispatch validation

Estimated Duration

2 Weeks

Epic 7 – Supplier & Purchase Order Management

Objective

Manage suppliers and procurement workflows.

Features

- Supplier management
- Purchase order creation
- Approval workflow
- PO status tracking

Estimated Duration

2 Weeks

Epic 8 – Reporting & Dashboard

Objective

Provide operational visibility and analytics.

Features

- Inventory summary dashboard
- Low stock reports
- Stock movement reports
- Inventory valuation
- Export to Excel/PDF

Estimated Duration

2 Weeks

Epic 9 – Alerts & Notifications

Objective

Notify users about critical inventory events.

Features

- Low stock alerts
- Expiry alerts
- Email notifications
- In-app notifications

Estimated Duration

1 Week

Epic 10 – QA, Performance & Security Hardening

Objective

Ensure system quality and production readiness.

Features

- Functional testing
- Integration testing
- Performance optimization
- Security testing
- Bug fixing
- Regression testing

Estimated Duration

3 Weeks

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6. Suggested Sprint Plan

Sprint 0 (Week 1–2)

Focus

Planning & Foundation

Deliverables

- Requirements finalized
- Architecture completed
- Dev environments ready
- Repository setup

- CI/CD setup

Sprint 1 (Week 3–4)

Focus

Authentication + Base UI

Deliverables

- Login/logout
- RBAC
- User management
- Base frontend layout

Sprint 2 (Week 5–6)

Focus

Product Management

Deliverables

- Product CRUD
- Categories
- SKU management
- Product search

Sprint 3 (Week 7–8)

Focus

Inventory Management

Deliverables

- Stock tracking
- Warehouse support
- Inventory ledger

Sprint 4 (Week 9–10)

Focus

Stock Inward & Supplier Management

Deliverables

- Goods receiving
- Supplier module
- Purchase orders

Sprint 5 (Week 11–12)

Focus

Stock Outward & Transfers

Deliverables

- Dispatch workflow
- Returns
- Transfers

Sprint 6 (Week 13–14)

Focus

Reporting & Notifications

Deliverables

- Dashboard
- Reports
- Alerts

Sprint 7 (Week 15–16)

Focus

System Integration & Bug Fixes

Deliverables

- Integrated workflows
- Major defect fixes

Sprint 8 (Week 17–18)

Focus

QA & UAT

Deliverables

- UAT testing
- Performance testing
- Security validation

Sprint 9 (Week 19–20)

Focus

Deployment & Hypercare

Deliverables

- Production deployment
- User training
- Go-live support

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7. Major Milestones

Milestone	Timeline
Requirements Signoff	End of Week 2
Architecture Finalized	End of Week 2
MVP Ready	End of Week 10
Feature Complete	End of Week 14
QA Complete	End of Week 18
UAT Signoff	End of Week 18
Production Go-Live	Week 20

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8. Recommended Developer Allocation

Area	Primary Owner
Backend APIs	Backend Developer
Frontend UI	Frontend Developer
Database & Architecture	Tech Lead
Integrations & Shared Features	Full Stack Developer

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9. Risks & Mitigation

Risk	Mitigation
Scope expansion	Freeze MVP scope
Delayed approvals	Weekly stakeholder demos
Inventory accuracy issues	Strong audit logs & validations
Performance bottlenecks	Early load testing
Security vulnerabilities	Security reviews each sprint

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10. MVP Recommendation

Phase 1 (Must Have)

- Authentication
- Product management
- Inventory tracking
- Stock inward/outward
- Reporting
- Audit logs

Phase 2 (Nice to Have)

- Barcode scanning
- Mobile support
- ERP integrations
- AI forecasting
- RFID support

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11. Suggested Success Metrics

Metric	Target
Inventory accuracy	>98%
System uptime	99.9%
Response time	<3 seconds
UAT approval	100%
Critical defects in production	Minimal/Zero